

NORMAL DISTRIBUTION

1. What is Normal Distribution?

A **Normal Distribution** (also called the **Gaussian Distribution** or **Bell Curve**) is one of the most important probability distributions in statistics and machine learning.

It is a **continuous probability distribution** where most observations are concentrated around the center (mean), and fewer observations occur as we move away from the center.

Simple Definition

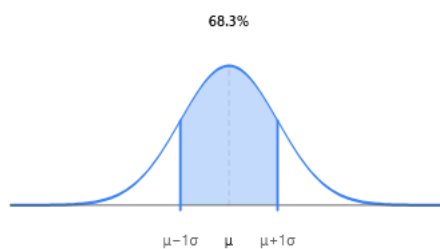
A Normal Distribution is a bell-shaped probability distribution in which most values lie near the mean, and the probability of values decreases equally on both sides of the mean.

Why is it Called a Bell Curve?

The graph of a normal distribution looks like a bell.

- Highest point at the center
- Left and right sides are perfectly symmetrical
- Long tails on both sides

68.3% $\mu - 1\sigma$ μ $\mu + 1\sigma$



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Real-Life Examples

Many natural and real-world phenomena approximately follow a normal distribution:

Example	Normally Distributed?
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Student Heights	✓ Yes
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Example	Normally Distributed?
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IQ Scores	✓ Yes
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Blood Pressure	✓ Yes
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Human Weight	Approximately
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Measurement Errors	✓ Yes
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Exam Scores (Large Classes)	Often Yes
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Manufacturing Dimensions	Often Yes
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Why is Normal Distribution Important?

Normal distribution is the foundation of

- Statistics
- Data Science
- Machine Learning
- Artificial Intelligence
- Deep Learning

Many statistical methods assume that the data follows a normal distribution.

Why is Normal Distribution Used in Machine Learning?

Many Machine Learning algorithms perform better when features are approximately normally distributed.

Normal distribution helps in:

- Data preprocessing
- Feature scaling
- Outlier detection
- Probability estimation
- Statistical testing
- Model assumptions

PROPERTIES OF NORMAL DISTRIBUTION

Property	Description
Shape	Bell-shaped
Symmetry	Perfectly symmetric
Mean	Center
Median	Equals mean
Mode	Equals mean
Area	Total area = 1
Skewness	0
Kurtosis	Approximately 3 (mesokurtic)

STANDARD NORMAL DISTRIBUTION

A standard normal distribution is a special case where

Mean = 0

Standard Deviation = 1